

# Six years, ~73,000 trades, two universes — and an honest read on what it proves

The tuned strategist is profitable and statistically well-separated from chance across both large-cap and small-cap universes. The binding constraint on confidence isn't sample size — it's that option premiums are modeled, not quoted.

**Window** Oct 2020 – Jul 2026    **Universe** 500 S&P + 127 small-cap    **Tenor** 35 trading days, rolled every 10  
**Report** 10–11 Jul 2026 (incl. overnight validation suite)

## Headline result

- Post-review: real structural alpha · tail-event profile · VRP-dependent

S&P 500 — RETURN ON RISK

**+21.5%**

52.6% win rate over 60,954 trades. Every regime bucket positive, including a bear regime that now returns **+17.2%**.

SMALL-CAP — RETURN ON RISK

**+7.9%**

61.4% win rate over 11,985 trades. Weaker in absolute terms, but well-calibrated and positive in every regime.

## Performance by structure

Return on risk = mean P&L as a fraction of the capital at risk per trade — the unit that's comparable across a \$60 stock and a \$600 one. Win rate and profitability diverge by design: directional debit spreads win under half the time but pay more on wins than they lose.

| S&P 500 STRUCTURE     | TRADES        | WIN %        | PRED POP     | RETURN / RISK |
|-----------------------|---------------|--------------|--------------|---------------|
| Long Straddle         | 25,558        | 50.9%        | 42.7%        | +24.4%        |
| Call Debit Spread     | 29,881        | 49.2%        | 42.1%        | +21.2%        |
| Bull Put Spread       | 1,695         | 81.8%        | 69.4%        | +16.7%        |
| Iron Condor           | 1,732         | 77.9%        | 63.2%        | +16.6%        |
| Bear Call Spread      | 62            | 91.9%        | 81.8%        | +11.8%        |
| Cash-Secured Put      | 1,712         | 84.6%        | 72.5%        | +2.2%         |
| Put Debit Spread      | 314           | 31.5%        | 44.5%        | -27.0%        |
| <b>All structures</b> | <b>60,954</b> | <b>52.6%</b> | <b>44.6%</b> | <b>+21.5%</b> |

Put Debit Spread (−27%) is the one known residual: 314 trades, 0.5% of volume — negligible drag, deliberately not chased to avoid over-fitting.

| SMALL-CAP STRUCTURE   | TRADES        | WIN %        | PRED POP     | RETURN / RISK |
|-----------------------|---------------|--------------|--------------|---------------|
| Long Straddle         | 2,827         | 44.1%        | 42.4%        | +17.9%        |
| Iron Condor           | 793           | 74.7%        | 63.4%        | +14.6%        |
| Cash-Secured Put      | 3,226         | 74.2%        | 69.6%        | +6.9%         |
| Bull Put Spread       | 2,891         | 68.5%        | 66.5%        | +3.5%         |
| Bear Call Spread      | 372           | 83.1%        | 81.4%        | +2.4%         |
| Put Debit Spread      | 1,876         | 44.6%        | 46.2%        | -0.4%         |
| <b>All structures</b> | <b>11,985</b> | <b>61.4%</b> | <b>58.7%</b> | <b>+7.9%</b>  |

Small-cap preset favours selling premium (CSP, credit spreads) over buying direction — richer premium, no reliance on a steady upward grind that small-caps don't have.

## What the tuning bought (S&P)

Three rounds of changes, each held to the same 6-year backtest. The bear-regime line is the standout — it flipped from a persistent loss to the second-strongest regime.

|                           |             |        |
|---------------------------|-------------|--------|
| Regime gate only          | <div></div> | +14.4% |
| + 3 core levers           | <div></div> | +19.8% |
| + bear gate & size preset | <div></div> | +21.5% |

**Bear regime, before → after: -7.6% → +17.2%.** High-conviction bearish trades turned out to be *anti-predictive* in a bear tape — the strongest downtrends snap back hardest — so they now route to neutral structures, which earned +11–27% in exactly those windows.

## Is the edge real, or sampling noise?

Win rates below, with 95% intervals computed on an **overlap-adjusted** sample. Because a 35-day trade is re-entered every 10 days, consecutive trades on a name overlap  $\sim 3.5\times$ , so the effective independent count is roughly  $N \div 3.5$  — a deliberately conservative haircut. The  $\sigma$  column is standard deviations above a coin-flip.

| BUCKET            | TRADES | EFF. N | WIN % | 95% INTERVAL | $\Sigma$ VS 50% |
|-------------------|--------|--------|-------|--------------|-----------------|
| S&P — all         | 60,954 | 17,415 | 52.6% | 51.9 – 53.3% | +6.9            |
| S&P — bear regime | 4,020  | 1,149  | 56.0% | 53.1 – 58.8% | +4.1            |
| Small-cap — all   | 11,985 | 3,424  | 61.4% | 59.8 – 63.0% | +13.7           |
| Iron Condor (S&P) | 1,732  | 495    | 77.9% | 74.0 – 81.3% | +15.0           |
| Bull Put (S&P)    | 1,695  | 484    | 81.8% | 78.1 – 85.0% | +18.1           |
| Call Debit (S&P)  | 29,881 | 8,537  | 49.2% | 48.1 – 50.3% | -1.5            |

Call Debit's win rate straddles 50% — expected, and not a weakness: its edge lives in payoff asymmetry (wins bigger than losses), which is why it still returns +21% on risk. Win rate is the wrong lens for directional structures.

**Verdict on significance:** every aggregate and every premium-selling structure clears  $4\sigma$  even after the overlap haircut — the win-rate edge is not sampling noise. What that does *not* settle is whether the edge survives real fills.

# Profit factor & risk-adjusted return

Win rate ignores *magnitude*. These metrics don't. Computed on 17,923 per-trade returns from a representative 150-name S&P subset — whose mean return-on-risk of +21.7% reproduces the full-universe +21.5%, confirming it's not cherry-picked. The 95% intervals come from a **symbol-block bootstrap** (resampling whole names, 2,000 draws) — the honest way to price in the overlap correlation, stronger than the win-rate haircut above.

MEAN RETURN ON RISK

+21.7%

95% CI [+19.2%, +24.1%]

PROFIT FACTOR

1.62

95% CI [1.54, 1.70]

SHARPE · PER TRADE

0.20

95% CI [0.18, 0.22]

**Reading these:** a profit factor of 1.62 means gross winners out-earn gross losers by ~62% — comfortably above the 1.0 breakeven, with the whole interval clear of it. The bootstrap interval on return also sits entirely above zero, so it's now the *profitability*, not just the win rate, that's statistically separated from noise. Per-trade Sharpe annualizes to roughly 1.0 — respectable — but overlap makes that scaling optimistic, so treat it as a ceiling, not a promise.

| STRUCTURE (S&P SUBSET) | TRADES | PROFIT FACTOR | SHARPE / TRADE | MEAN R |
|------------------------|--------|---------------|----------------|--------|
| Cash-Secured Put       | 459    | 2.59          | 0.362          | +2.1%  |
| Long Straddle          | 7,661  | 1.98          | 0.247          | +24.6% |
| Bull Put Spread        | 453    | 1.92          | 0.292          | +14.0% |
| Iron Condor            | 450    | 1.77          | 0.264          | +14.8% |
| Call Debit Spread      | 8,869  | 1.45          | 0.173          | +21.0% |

Cash-Secured Put shows the paradox cleanly: highest profit factor and Sharpe (small, reliable wins) but the lowest return on risk (huge collateral). Long Straddle is the standout — high magnitude *and* high consistency.

# Validation suite: ten acceptance gates

A professional-grade gate list (walk-forward, baselines, plateaus, robustness, drawdown) run against the tuned strategy. Every gate that *can* be tested on synthetic data has been; what remains needs real quotes and time.

| ACCEPTANCE GATE                          |                                                                       |
|------------------------------------------|-----------------------------------------------------------------------|
| Expectancy positive (validation window)  | 2024–26 is <i>validation</i> , not test — all rules built with the fu |
| Lower confidence bound > 0               | 2024+ only, <b>net of costs</b> , entry-date-cohort bootstrap: m      |
| Survives 2× costs                        | Breakeven at ~11% round-trip slip; +11.5% at 5%                       |
| Parameter plateaus, not cliffs           | Full 3×3 DTE×delta joint grid: all 9 cells +19.5–25.5% in bc          |
| No concentration                         | Straddles 48.5% of P&L; <b>without best 5% of trades: +0.0</b>        |
| Retention across windows                 | 110% (S&P), 87% (small-cap) — but can't detect pricing b              |
| Drawdown fits risk budget                | <b>Daily-marked MDD –21.2%</b> at 5% risk cap (1.6× the exp           |
| Beats baselines (incl. permuted signals) | Beats blind condor and buy-and-hold; <b>ties permuted-sig</b>         |
| Not merely equity beta                   | Daily HAC regression: <b>alpha +12.2 bps/day (+31%/yr gr</b>          |
| Paper fills ≈ modeled fills              | Recorder (109 symbols) + daily signal log (legs vs bid/ask,           |

## The identity finding: the signal is a passenger

The decisive test of the review round: rerun the entire backtest with each symbol's technical decisions *shuffled across its own entry dates* — same chains, same gates, same mechanics, destroyed signal. The result is indistinguishable from the real run.

|                           | MEAN R  | PF   | WIN % |
|---------------------------|---------|------|-------|
| Real technical signals    | +21.7%  | 1.62 | 52.6% |
| Randomly permuted signals | +21.67% | 1.63 | 52.4% |

Corroborated by the ranking test: POP deciles rank outcomes cleanly end-to-end (pred 38→65% maps to actual 45→75%, strictly monotone), but *within-structure* ranking ICs are ~0.02–0.05 — negligible. The strategist's real skill is coarse: vol-regime structure routing and the gates. The directional technical layer adds ≈ nothing to aggregate P&L.

What this strategy actually is: a mechanical **long-volatility / VRP tail-harvester** with regime and EV gating. Its P&L level rides on the volatility-risk-premium assumption (blind ATM straddles earn +19% at the assumed IV = 1.05× realized, break even at 1.25×, lose at 1.40×) — and the first real-quote reading (25 names, day one of the recorder) measured **median actual VRP ≈ 0.99**: the market currently sells that convexity *cheaper* than the backtest assumed. Weeks of recorder data will settle the strategy's single load-bearing assumption.

## Walk-forward: does the edge survive out-of-sample?

The levers were developed on 2020–2023 patterns. Frozen and applied to 2024–2026 — data no lever value ever saw — the strategy performs *better*, and each lever passes the harder test: it would have been adopted on in-sample data alone *and* still adds out-of-sample.

| CONFIG             | IS MEAN R | IS PF | OOS MEAN R | OOS PF | RETENTION        |
|--------------------|-----------|-------|------------|--------|------------------|
| FINAL (S&P)        | +20.7%    | 1.57  | +22.8%     | 1.67   | 110%             |
| FINAL (small-cap)  | +9.4%     | 1.40  | +8.2%      | 1.36   | 87%              |
| without EV gate    | +17.4%    | 1.48  | +20.3%     | 1.59   | lever +2.5pt OOS |
| without drift      | +14.5%    | 1.48  | +15.9%     | 1.51   | lever +6.9pt OOS |
| without bear gates | +14.5%    | 1.36  | +20.8%     | 1.60   | lever +2.0pt OOS |

## Baseline: is the tuning worth anything?

Two mechanical comparators on the same chains, same dates, no signals. The raw return of "always buy an ATM call debit" nearly matches the tuned strategy — but decomposition shows it's ~80% levered market exposure. The tuned strategist earns **3× the alpha at 40% of the beta**; the mechanical premium-seller loses outright.

|                         | MEAN R        | PF          | SHARPE/TRADE | ALPHA         | BETA       |
|-------------------------|---------------|-------------|--------------|---------------|------------|
| <b>Tuned strategist</b> | <b>+21.7%</b> | <b>1.62</b> | <b>0.200</b> | <b>+14.6%</b> | <b>3.3</b> |
| Always-buy call debit   | +21.0%        | 1.41        | 0.160        | +4.5%         | 7.8        |
| Always-sell iron condor | -6.7%         | 0.81        | —            | —             | —          |

## Costs, plateaus, robustness, drawdown

- **Cost stress:** at 2.5% round-trip slippage PF 1.42 / +16.0%; at 5% PF 1.29 / +11.5%; breakeven ≈ 11%; at 15% the strategy loses. The edge survives realistic costs but not sloppy execution — liquid, tight-spread underlyings only.
- **Parameter plateau:** every neighbor of the tuned point (drift 4/12, delta .25/.35, DTE 25/45) earns +20.1% to +23.4% — a smooth plateau, no curve-fit cliff. DTE 45 tested mildly better; left alone on the plateau principle.
- **Robustness:** no year supplies more than 31% of P&L, top symbol 1.9%, top five 9.0%. Honest flags: Long Straddle contributes 48.5% of P&L and the top 5% of trades supply 55% — a long-vol, tail-driven profit engine that deserves its own ask-side-fill validation before real money.
- **Drawdown (daily marks, per review):** marking every open position daily deepens MDD to **-21.2%** at the 5% risk cap (the expiry-mark -13.5% was a floor, as suspected); 388 days underwater; daily-marked portfolio Sharpe 1.47 gross ≈ 0.8 after realistic costs.
- **Honest portfolio Sharpe:** 0.57 on non-overlapping cohorts (the 1.47 overlapping figure is the optimistic bound).

## Confidence, along four axes

A single number would flatter the result. Confidence is high on some dimensions and genuinely limited on others — and the limited ones are structural, not fixable with more data.

### Statistical significance

HIGH 

60,954 S&P trades; win rate holds at  $+6.9\sigma$  after a conservative overlap haircut, and a symbol-block bootstrap puts the return-on-risk interval at  $[+19.2\%, +24.1\%]$  — profitability itself, not just win rate, is clear of zero.

### Over-fitting risk

VERY LOW 

Now evidenced, not just argued: 110% out-of-sample retention, every lever valid on in-sample data alone, and a smooth parameter plateau (all neighbors  $+20\text{--}23\%$ ). Curve-fit strategies do not do this.

### Regime coverage

MODERATE 

~93% of trades fall in bull regimes. The bull result is deeply sampled; the bear result (~4k trades) is positive and significant but rests on one 2022-era stretch — thinner evidence.

### Transfer to live P&L

LIMITED, NOW BOUNDED 

Premiums are still modeled, but the cost stress bounds the exposure: the edge survives 5% round-trip slippage ( $+11.5\%$ ) and dies at  $\sim 11\%$ . A daily recorder now archives real chains (109 symbols, sector-stratified) — the path to closing this axis with paper-fill comparison.

## What this report does *not* prove

The single biggest caveat: **no historical option chains exist for this account**, so entry premiums are priced by Black-Scholes off trailing realized volatility. Any edge that lives in **real implied vol vs realized** — a large part of live options P&L — is assumed away here.

- **No transaction costs.** Commissions, bid-ask slippage, and early-assignment risk are unmodeled; real net returns sit below these figures.
- **Hold-to-expiry marks.** The profit-taking and stop rules aren't simulated — conservative for premium sellers, generous for debit buyers.
- **Bull-heavy sample.** The window is dominated by a rising market; sustained-bear behaviour is thinly tested.
- **In-sample.** Parameters were chosen on this same 6-year window — low over-fit risk given how few there are, but not a walk-forward out-of-sample test.

## Bottom line

After a full external-review round, the honest identity is settled: this is a **mechanical long-volatility harvester** — disciplined structure routing off the vol regime, hard gates, and cheap convexity — whose alpha is real (+12.2 bps/day,  $t=2.99$ , on daily marks) but *structural*, not predictive: permuting the technical signals reproduces the result exactly. It is a **tail-event strategy** (breakeven without its best 5% of trades) whose level rides on one assumption — the volatility risk premium — that the daily recorder is now measuring against reality (first reading: favorable). Sizing doctrine: expect daily-marked drawdowns near **4× your standing risk cap**, year-long droughts, and profits arriving in bursts. Pilot at ~0.2% of capital at risk; judge it on years, not quarters. Nothing left to compute — from here, only real quotes and time raise confidence.

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Generated from runs saved at `data_store/reports/backtest_FINAL_sp500_2026-07-10.txt` and ...  
`_smallcap_2026-07-10.txt`. Return-on-risk and win rates as reported by the synthetic backtest. Win-rate intervals are Wilson score intervals on the overlap-adjusted effective sample (haircut 3.5×); profit factor, Sharpe, and

return intervals are symbol-block bootstraps (2,000 resamples of 17,923 per-trade returns across 150 names).  
Methodology caveats above bound all figures.